If two letters are in the same row then replace with Immediate letter

PT = . cd

CT . dg

p	l	a	y	f
i/j	r	e	x	m
b	c	d	g	h
k	n	o	q	s
t	u	v	w	z

p	l	a	y	f
i/j	r	e	x	m
b	c	d	g	h
k	n	o	q	s
t	u	v	w	z

* Plain text :- Computer

key :- security

⇒ co / mp / ut / er

S	E	C	U	R
I/J	T	Y	A	B
D	F	G	H	K
L	M	N	O	P
Q	U	W	X	Z

Co - UN
mp - NL
Ur - AE
ER - CS

1. A feistel cipher uses 8-bit block and one round.

   plaintext = 10101101

   Round key k_1 4100

   Round function: $F(R,k) = R \oplus k$

Question:-

- split PT to L_0 & R_0
- Compute $F(R_0, k_1)$
- Recover R_0 using Feistel property.
- write recovered PT block

2. An 8-bit PT is encrypted using two feistel rounds.

- PT = 11100010
- $k_1 = 0110$
- $k_2 = 1001$

3. Decryption — 00111100 One round feistel cipher, $k_1 = 0101$, $F(R,k) = R \oplus k$

- Split (Ciphertext) into L_1 & R_1
- Recover R_0 using feistel property
- Compute $F(R_0, k_1)$
- Recover original L_0
- Write recovered PT block.

(2) Two-Round Feistel cipher

$F(R,k) = R \oplus k$

PT = 11100010

k1 = 0110

k2 = 1001

Split PT = $L_0 = 1110$

$R_0 = 0010$

R1 ($k1 = 0110$)

$F(R_0, k_1) = 0010 \oplus 0110 = 0100$

$$\begin{array}{r} 0010 \\ 0110 \\ \hline 0100 \end{array}$$

$L_1 = R_0 = 0010$ $\boxed{(R_0, k_1) = 0100}$ new left

$R_1 = L_0 \oplus F = 1110 \oplus 0100 = \boxed{1010}$ R_1

After R1 =

$L_1 R_1 = 0010\ 1010$

R_2 ($k2 = 1001$) $\rightarrow R_2$

$F(R_1, k_2) = 1010 \oplus 1001 = 0011$

$L_2 = R_1 = 1010$

$R_2 = L_1 \oplus F = 0010 \oplus 0011 = 0001$

after R2

$L_2 R_2 = 1010\ 0001$

L(new) = R(old)

R(new) = L(old) XOR F(R(old), key)

$\boxed{CT = 10100001}$

hill cipher

PT = H | E | L | L | O
7 | 4 | 11 | 11 | 14

$k = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix}$

$= 15 - 6 = 9 \% 26$

key = 2 x 2

PT - H E | L L | O X

HE LL OX

En :- (i) (7, 4) (H E)

$$\begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 7 \\ 4 \end{pmatrix} = \begin{bmatrix} 33 \\ 34 \end{bmatrix}$$

Mod 26

33 mod 26 = 7 → H

34 mod 26 = 8 → I

(ii) (11, 11) (L, L)

$$\begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 11 \\ 11 \end{pmatrix} = \begin{bmatrix} 66 \\ 77 \end{bmatrix}$$

66 mod 26 = 14 → O

77 mod 26 = 23 → Z

$\boxed{\begin{matrix} \text{HELLOX} \\ \text{HIOZHN} \end{matrix}}$

(iii) (14, 23) (O, X)

$$\begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 14 \\ 23 \end{pmatrix} = \begin{bmatrix} 111 \\ 143 \end{bmatrix}$$

111 mod 26 = 7 → H

143 mod 26 = 13 → N

Decryption

mod 26

CT = H I O Z H N

$9 \times 1 = 9$
$9 \times 2 =$
$9 \times 3 = 27$
$27 - 26$
$= 1$

$key = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$

for finding inverse

$K = (3 \times 5) - (3 \times 2) = 9$

inverse of 9 module 26 :

$9^{-1} = 3$ =

Adj matrix = $\begin{bmatrix} 5 & -3 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 23 \\ 24 & 3 \end{bmatrix}$

inverse key :-

$K^{-1} = 3 \begin{bmatrix} 5 & 23 \\ 24 & 3 \end{bmatrix} \pmod{26} = \begin{bmatrix} 25 & 17 \\ 20 & 9 \end{bmatrix}$

Decrypt each pair

HI

$\begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 241 \\ 212 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$ mod 26

= HE

OZ

$\begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 14 \\ 25 \end{bmatrix} = \begin{bmatrix} 635 \\ 505 \end{bmatrix} = \begin{bmatrix} 11 \\ 11 \end{bmatrix}$ mod 26

= LL

HN

$$\begin{bmatrix}15 & 17\\ 20 & 9\end{bmatrix}\begin{bmatrix}7\\ 13\end{bmatrix} = \begin{bmatrix}326\\ 257\end{bmatrix} = \begin{bmatrix}14\\ 23\end{bmatrix} \bmod 26$$

$= OX$

HELLOX

$= HELLO$